

Random vibrations of coupled continuous systems

MG3416–Advanced Structural Acoustics - Lecture #5

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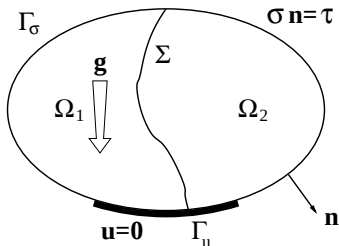
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General case

Notations and setting



- The system occupying the domain Ω is divided into two sub-systems occupying the domains Ω_1 and Ω_2 .
- Their interface is $\Sigma = \partial\Omega_1 \cap \partial\Omega_2$ with $|\Sigma| \neq 0$.
- The displacement field in each sub-system $r = 1$ or 2 is $\mathbf{u}_r = \mathbf{u}|_{\Omega_r}$, where $\mathbf{u}$ is the displacement field in Ω .

General case

Notations and setting

- They satisfy the following system derived from the VPP: Find $\mathbf{u}_r \in \mathcal{C}_r$, $r \in \{1, 2\}$, s.t.

$$\begin{bmatrix} \mathbf{M}_1 & \mathbf{M}_{12} \\ \mathbf{M}_{21} & \mathbf{M}_2 \end{bmatrix} \begin{pmatrix} \ddot{\mathbf{u}}_1 \\ \ddot{\mathbf{u}}_2 \end{pmatrix} + \begin{bmatrix} \mathbf{D}_1 & \mathbf{D}_{12} \\ \mathbf{D}_{21} & \mathbf{D}_2 \end{bmatrix} \begin{pmatrix} \dot{\mathbf{u}}_1 \\ \dot{\mathbf{u}}_2 \end{pmatrix} + \begin{bmatrix} \mathbf{K}_1 & \mathbf{K}_{12} \\ \mathbf{K}_{21} & \mathbf{K}_2 \end{bmatrix} \begin{pmatrix} \mathbf{u}_1 \\ \mathbf{u}_2 \end{pmatrix} = \begin{pmatrix} \mathbf{f}_1 \\ \mathbf{f}_2 \end{pmatrix},$$

where the sets of admissible displacement fields $\mathcal{C}_r$ are:

$$\mathcal{C}_r \subseteq \{ \mathbf{v}; \mathbf{v} \in L^2(\Omega_r), \nabla \mathbf{v} \in L^2(\Omega_r), \text{ and } \mathbf{v}|_{\Gamma_u \cap \partial\Omega_r} = \mathbf{0} \}.$$

- **Hypotheses:** the coupling operators are s.t.
- The mass coupling:

$$m_c \begin{pmatrix} \mathbf{u}_1 | \mathbf{v}_1 \\ \mathbf{u}_2 | \mathbf{v}_2 \end{pmatrix} = \langle \mathbf{M}_{12} \mathbf{u}_2, \mathbf{v}_1 \rangle_{\mathcal{C}'_1, \mathcal{C}_1} + \langle \mathbf{M}_{21} \mathbf{u}_1, \mathbf{v}_2 \rangle_{\mathcal{C}'_2, \mathcal{C}_2}$$

is symmetric $\mathbf{M}_{12} \equiv \mathbf{M}_{21}^\top$;

General case

Notations and setting

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$$\begin{bmatrix} \mathbf{M}_1 & \mathbf{M}_{12} \\ \mathbf{M}_{21} & \mathbf{M}_2 \end{bmatrix} \begin{pmatrix} \ddot{\mathbf{u}}_1 \\ \ddot{\mathbf{u}}_2 \end{pmatrix} + \begin{bmatrix} \mathbf{D}_1 & \mathbf{D}_{12} \\ \mathbf{D}_{21} & \mathbf{D}_2 \end{bmatrix} \begin{pmatrix} \dot{\mathbf{u}}_1 \\ \dot{\mathbf{u}}_2 \end{pmatrix} + \begin{bmatrix} \mathbf{K}_1 & \mathbf{K}_{12} \\ \mathbf{K}_{21} & \mathbf{K}_2 \end{bmatrix} \begin{pmatrix} \mathbf{u}_1 \\ \mathbf{u}_2 \end{pmatrix} = \begin{pmatrix} \mathbf{f}_1 \\ \mathbf{f}_2 \end{pmatrix},$$

where the sets of admissible displacement fields $\mathcal{C}_r$ are:

$$\mathcal{C}_r \subseteq \{ \mathbf{v}; \mathbf{v} \in L^2(\Omega_r), \nabla \mathbf{v} \in L^2(\Omega_r), \text{ and } \mathbf{v}|_{\Gamma_u \cap \partial\Omega_r} = \mathbf{0} \}.$$

- **Hypotheses:** the coupling operators are s.t.
- The stiffness coupling:

$$k_c \begin{pmatrix} \mathbf{u}_1 | \mathbf{v}_1 \\ \mathbf{u}_2 | \mathbf{v}_2 \end{pmatrix} = \langle \mathbf{K}_{12} \mathbf{u}_2, \mathbf{v}_1 \rangle_{\mathcal{C}'_1, \mathcal{C}_1} + \langle \mathbf{K}_{21} \mathbf{u}_1, \mathbf{v}_2 \rangle_{\mathcal{C}'_2, \mathcal{C}_2}$$

is symmetric $\mathbf{K}_{12} \equiv \mathbf{K}_{21}^\top$;

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Notations and setting

- They satisfy the following system derived from the VPP: Find $\mathbf{u}_r \in \mathcal{C}_r$, $r \in \{1, 2\}$, s.t.

$$\begin{bmatrix} \mathbf{M}_1 & \mathbf{M}_{12} \\ \mathbf{M}_{21} & \mathbf{M}_2 \end{bmatrix} \begin{pmatrix} \ddot{\mathbf{u}}_1 \\ \ddot{\mathbf{u}}_2 \end{pmatrix} + \begin{bmatrix} \mathbf{D}_1 & \mathbf{D}_{12} \\ \mathbf{D}_{21} & \mathbf{D}_2 \end{bmatrix} \begin{pmatrix} \dot{\mathbf{u}}_1 \\ \dot{\mathbf{u}}_2 \end{pmatrix} + \begin{bmatrix} \mathbf{K}_1 & \mathbf{K}_{12} \\ \mathbf{K}_{21} & \mathbf{K}_2 \end{bmatrix} \begin{pmatrix} \mathbf{u}_1 \\ \mathbf{u}_2 \end{pmatrix} = \begin{pmatrix} \mathbf{f}_1 \\ \mathbf{f}_2 \end{pmatrix},$$

where the sets of admissible displacement fields $\mathcal{C}_r$ are:

$$\mathcal{C}_r \subseteq \{ \mathbf{v}; \mathbf{v} \in L^2(\Omega_r), \nabla \mathbf{v} \in L^2(\Omega_r), \text{ and } \mathbf{v}|_{\Gamma_u \cap \partial\Omega_r} = \mathbf{0} \}.$$

- **Hypotheses:** the coupling operators are s.t.
- The gyroscopic coupling:

$$d_c \left(\begin{matrix} \mathbf{u}_1 \\ \mathbf{u}_2 \end{matrix} \middle| \begin{matrix} \mathbf{v}_1 \\ \mathbf{v}_2 \end{matrix} \right) = \langle \mathbf{D}_{12} \mathbf{u}_2, \mathbf{v}_1 \rangle_{\mathcal{C}'_1, \mathcal{C}_1} + \langle \mathbf{D}_{21} \mathbf{u}_1, \mathbf{v}_2 \rangle_{\mathcal{C}'_2, \mathcal{C}_2}$$

is skew-symmetric $\mathbf{D}_{12} \equiv -\mathbf{D}_{21}^\top$.

General case

Eigenmodes and modal expansion

- **Spectral problem:** Find $\lambda \in \mathbb{R}$ and $\phi \in \mathcal{C}_r^b$ s.t.

$$\mathbf{K}_r \phi = \lambda \mathbf{M}_r \phi,$$

where $\mathcal{C}_r^b \subset L^2(\Omega_r)$ is the set of admissible displacement fields in Ω_r when some boundary conditions are prescribed on the interface Σ .

- **Examples:**
 - Craig-Bampton method (1968): $\phi|_{\Sigma} = \mathbf{0}$;
 - McNeal method (1971): $\sigma_r(\phi) \mathbf{n}_r = \mathbf{0}$ on Σ ;
 - Gladwell method (1964): $\sigma_r(\phi) \mathbf{n}_r = \lambda_r \mathcal{M} \phi|_{\Sigma}$ on Σ , for some $\mathcal{M} > 0$.
- **Remark:** *this issue is marginally addressed in the SEA literature, because the choice of $\mathcal{C}_r^b$ has only a mild influence on the method.*

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- **Hypothesis:** it is assumed that this problem admits a countable set of solutions $(\lambda_{r1}, \phi_{r1}), (\lambda_{r2}, \phi_{r2}) \dots$ s.t. $0 < \lambda_{r1} \leq \lambda_{r2} \leq \dots$ and $\{\phi_{r\alpha}\}_{\alpha \in \mathbb{N}^*}$ is an Hilbertian basis of the space $H_r = L^2_\mu(\Omega_r)$ of square integrable functions with respect to the unit mass measure $\mu_r(d\mathbf{x}) = \mathbb{1}_{\Omega_r} \frac{\varrho(d\mathbf{x})}{M_r}$, with $M_r = \int_{\Omega_r} \varrho d\mathbf{x}$:

$$\langle \mathbf{M}_r \phi_{r\alpha}, \phi_{r\alpha'} \rangle_{\mathcal{C}'_r, \mathcal{C}_r} = M_r \delta_{\alpha\alpha'},$$

$$\langle \mathbf{K}_r \phi_{r\alpha}, \phi_{r\alpha'} \rangle_{\mathcal{C}'_r, \mathcal{C}_r} = M_r \omega_{r\alpha}^2 \delta_{\alpha\alpha'},$$

where $\omega_{r\alpha}^2 = \lambda_{r\alpha}$.

- **Notations:** $r, s \in \{1, 2\}$, $\alpha, \alpha' \in \mathbb{N}^*$ are generic modes for the r^{th} sub-system, and $\beta, \beta' \in \mathbb{N}^*$ are generic modes for the s^{th} sub-system.

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Eigenmodes and modal expansion

- Consequently, the solution $\mathbf{u}_r \in \mathcal{C}_r^b$ can be expanded on the eigenbasis $\{\phi_{r\alpha}\}_{\alpha \in \mathbb{N}^*}$ as:

$$\mathbf{u}_r(\mathbf{x}, t) = \sum_{\alpha=1}^{\infty} q_{r\alpha}(t) \phi_{r\alpha}(\mathbf{x}),$$

- Introducing $\mu_r(\mathbf{u}, \mathbf{v}) = \int_{\Omega} \mathbf{u} \cdot \mathbf{v} \mu_r(d\mathbf{x})$ (the scalar product in H_r), the generalized coordinates $\{q_{r\alpha}\}_{\alpha \in \mathbb{N}^*}$ are:

$$q_{r\alpha} = \mu_r(\mathbf{u}, \phi_{r\alpha}).$$

- The μ_r -norm $\|\mathbf{u}\|_{\mu_r} = \sqrt{\mu_r(\mathbf{u}, \mathbf{u})}$ is obtained as:

$$\|\mathbf{u}(\cdot, t)\|_{\mu_r} = \left(\sum_{\alpha=1}^{+\infty} (q_{r\alpha}(t))^2 \right)^{\frac{1}{2}} < +\infty.$$

General case

Eigenmodes and modal expansion

- **Hypothesis** (Basile): the eigenmodes $\{\phi_{r\alpha}\}_{\alpha \in \mathbb{N}^*}$ diagonalize the damping operator as well:

$$\langle \mathbf{D}_r \phi_{r\alpha}, \phi_{r\alpha'} \rangle_{\mathcal{C}'_r, \mathcal{C}_r} = M_r \eta_{r\alpha} \omega_{r\alpha} \delta_{\alpha\alpha'},$$

where for $r \in \{1, 2\}$:

- $\omega_{r\alpha}$: the "blocked" (angular) frequency of the α^{th} mode of the r^{th} sub-system,
- $\xi_{r\alpha}$: the modal critical damping rate of the α^{th} mode of the r^{th} sub-system,
- $\eta_{r\alpha} = 2\xi_{r\alpha}$: the modal loss factor of the α^{th} mode of the r^{th} sub-system.

General case

Eigenmodes and modal expansion

- Owing to the Basile hypothesis, the generalized coordinates $\{q_{r\alpha}\}_{\alpha \in \mathbb{N}^*}$ satisfy:

$$M_r(\ddot{q}_{r\alpha}(t) + \eta_{r\alpha}\omega_{r\alpha}\dot{q}_{r\alpha}(t) + \omega_{r\alpha}^2 q_{r\alpha}(t)) = f_{r\alpha}(t) - \sum_{\beta=1}^{+\infty} (\mu_{\alpha\beta}\ddot{q}_{s\beta}(t) + \gamma_{\alpha\beta}\dot{q}_{s\beta}(t) + \kappa_{\alpha\beta}q_{s\beta}(t)) , \quad t \in \mathbb{R} ,$$

with $q_{r\alpha}(0) = \mu_r(\mathbf{u}_0, \phi_{r\alpha})$, $\dot{q}_{r\alpha}(0) = \mu_r(\mathbf{v}_0, \phi_{r\alpha})$, and:

$$f_{r\alpha} = \langle \mathbf{f}_r, \phi_{r\alpha} \rangle_{\mathcal{C}'_r, \mathcal{C}_r} ,$$

$$\mu_{\alpha\beta} = \langle \mathbf{M}_{rs} \phi_{s\beta}, \phi_{r\alpha} \rangle_{\mathcal{C}'_r, \mathcal{C}_r} = \langle \mathbf{M}_{sr} \phi_{r\alpha}, \phi_{s\beta} \rangle_{\mathcal{C}'_s, \mathcal{C}_s} ,$$

$$\gamma_{\alpha\beta} = \langle \mathbf{D}_{rs} \phi_{s\beta}, \phi_{r\alpha} \rangle_{\mathcal{C}'_r, \mathcal{C}_r} = - \langle \mathbf{D}_{sr} \phi_{r\alpha}, \phi_{s\beta} \rangle_{\mathcal{C}'_s, \mathcal{C}_s} ,$$

$$\kappa_{\alpha\beta} = \langle \mathbf{K}_{rs} \phi_{s\beta}, \phi_{r\alpha} \rangle_{\mathcal{C}'_r, \mathcal{C}_r} = \langle \mathbf{K}_{sr} \phi_{r\alpha}, \phi_{s\beta} \rangle_{\mathcal{C}'_s, \mathcal{C}_s} .$$

Energetic quantities

Definitions

Definition

- The *mechanical energy* of the coupled system $r \in \{1, 2\}$:

$$\begin{aligned}\mathcal{E}(t) &= \mathcal{E}_1(t) + \mathcal{E}_2(t) \\ &+ \sum_{\alpha=1}^{+\infty} \sum_{\beta=1}^{+\infty} (\mu_{\alpha\beta} \dot{q}_{1\alpha}(t) \dot{q}_{2\beta}(t) + \kappa_{\alpha\beta} q_{1\alpha}(t) q_{2\beta}(t)) ,\end{aligned}$$

where $\mathcal{E}_r(t) = \frac{1}{2} M_r \sum_{\alpha=1}^{+\infty} [(\dot{q}_{r\alpha}(t))^2 + \omega_{r\alpha}^2 (q_{r\alpha}(t))^2]$.

- The *dissipated power*: $\Pi_d(t) = \Pi_{d1}(t) + \Pi_{d2}(t)$, where $\Pi_{dr}(t) = M_r \sum_{\alpha=1}^{+\infty} \eta_{r\alpha} \omega_{r\alpha} (\dot{q}_{r\alpha}(t))^2$.
- The *input power*: $\Pi_{IN}(t) = \Pi_{IN1}(t) + \Pi_{IN2}(t)$, where $\Pi_{INr}(t) = \sum_{\alpha=1}^{+\infty} f_{r\alpha}(t) \dot{q}_{r\alpha}(t)$.

Energetic quantities

Power flow between the sub-systems

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- The powers exchanged by the sub-systems:

$$\begin{aligned}\Pi_{12}(t) &= (\text{force } 1 \rightarrow 2) \times (\text{celerity } 2) \\ &= - \sum_{\alpha=1}^{+\infty} \sum_{\beta=1}^{+\infty} \dot{q}_{2\beta}(t) (\mu_{\alpha\beta} \ddot{q}_{1\alpha}(t) - \gamma_{\alpha\beta} \dot{q}_{1\alpha}(t) + \kappa_{\alpha\beta} q_{1\alpha}(t)) ;\end{aligned}$$

$$\begin{aligned}\Pi_{21}(t) &= (\text{force } 2 \rightarrow 1) \times (\text{celerity } 1) \\ &= - \sum_{\alpha=1}^{+\infty} \sum_{\beta=1}^{+\infty} \dot{q}_{1\alpha}(t) (\mu_{\alpha\beta} \ddot{q}_{2\beta}(t) + \gamma_{\alpha\beta} \dot{q}_{2\beta}(t) + \kappa_{\alpha\beta} q_{2\beta}(t)) .\end{aligned}$$

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- The instantaneous power of the overall system:

$$\Pi(t) = \dot{\mathcal{E}}(t) = \dot{\mathcal{E}}_1(t) + \dot{\mathcal{E}}_2(t) - \Pi_{12}(t) - \Pi_{21}(t),$$

- The instantaneous power balance reads:

$$\dot{\mathcal{E}}_1(t) + \dot{\mathcal{E}}_2(t) = \Pi_{\text{IN}}(t) - \Pi_{\text{d}}(t) + \Pi_{12}(t) + \Pi_{21}(t),$$

or for each sub-system:

$$\dot{\mathcal{E}}_r(t) = \Pi_{\text{IN}r}(t) - \Pi_{\text{d}r}(t) + \Pi_{sr}(t), \quad s \neq r \in \{1, 2\}.$$

Stationary excitations

Definition of the stationary loads

- **Data:** $(\mathbf{F}_{r,t}, t \in \mathbb{R}), r \in \{1, 2\}$, are second order, centered stochastic processes defined on $(\Omega, \mathcal{F}, P)$, indexed on $\mathbb{R}$, with values in $[L^2(\Omega_r)]^3$, and m.s.s.

- **Hypotheses:**

- $\exists \omega \mapsto \mathbf{S}_{\mathbf{F}_r}(\omega; \mathbf{x}, \mathbf{y}) : \mathbb{R} \rightarrow \mathbb{C}^{3 \times 3}, \mathbf{x}, \mathbf{y} \in \Omega_r$, Hermitian, positive, integrable on $\mathbb{R}_\omega$, and s.t.:

$$\mathbf{S}_{\mathbf{F}_r}(\omega; \mathbf{x}, \mathbf{y}) = \mathbf{S}_{\mathbf{F}_r}(\omega; \mathbf{y}, \mathbf{x})^*$$

$$\mathbf{S}_{\mathbf{F}_r}(-\omega; \mathbf{x}, \mathbf{y}) = \overline{\mathbf{S}_{\mathbf{F}_r}(\omega; \mathbf{x}, \mathbf{y})}$$

$$\mathbf{S}_{\mathbf{F}_r}(\omega; \mathbf{x}, \mathbf{y}) = \mathbf{S}_r(\mathbf{x}, \mathbf{y}) \otimes \mathbb{1}_{I_0 \cup \underline{I}_0}(\omega),$$

where:

$$I_0 \cup \underline{I}_0 = \left[\omega_0 - \frac{\Delta\omega}{2}, \omega_0 + \frac{\Delta\omega}{2} \right] \cup \left[-\omega_0 - \frac{\Delta\omega}{2}, -\omega_0 + \frac{\Delta\omega}{2} \right].$$

- $(\mathbf{F}_{1,t}, t \in \mathbb{R})$ and $(\mathbf{F}_{2,t}, t \in \mathbb{R})$ are **uncorrelated**:

$$\mathbb{E}\{\mathbf{F}_{1,t} \otimes \mathbf{F}_{2,t'}\} = \mathbf{0}, \quad \forall (t, t') \in \mathbb{R} \times \mathbb{R}.$$

Stationary excitations

Forced responses of the coupled system

- The sub-system forced responses $t \mapsto \mathbf{u}_r^f(\cdot, t)$ for $r \in \{1, 2\}$ are modeled by stochastic processes $(\mathbf{U}_{r,t}, t \in \mathbb{R})$ the properties of which are derived from filtering and mean-square derivation (see Lecture #1 part A).

Proposition

- $(\mathbf{U}_{r,t}, t \in \mathbb{R}), r \in \{1, 2\}$, are second order, centered stochastic processes defined on $(\Omega, \mathcal{F}, P)$, indexed on $\mathbb{R}$, with values in $\mathcal{C}_r^b$, and mean-square stationary.
- The same holds for their mean-square derivatives $(\dot{\mathbf{U}}_{r,t}, t \in \mathbb{R})$ and $(\ddot{\mathbf{U}}_{r,t}, t \in \mathbb{R})$, with values in H_r and $\mathcal{C}_r^{b'}$ respectively.

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Energetics of the stationary forced responses

- The instantaneous power balance reads:

$$\dot{\mathcal{E}}_{1,t} + \dot{\mathcal{E}}_{2,t} = \Pi_{\text{IN},t} - \Pi_{\text{d},t} + \Pi_{12,t} + \Pi_{21,t},$$

or for each sub-system:

$$\dot{\mathcal{E}}_{r,t} = \Pi_{\text{IN}r,t} - \Pi_{\text{dr},t} + \Pi_{sr,t}, \quad s \neq r \in \{1, 2\},$$

as equalities of second-order random variables.

- From the foregoing results, $\mathbb{E}\{\dot{\mathcal{E}}_{r,t}\} = 0$ for both sub-systems and $\mathbb{E}\{\Pi_{12,t}\} = -\mathbb{E}\{\Pi_{21,t}\}$; hence:

$$\begin{aligned}\mathbb{E}\{\Pi_{\text{IN},t}\} &= \mathbb{E}\{\Pi_{\text{d},t}\}, \\ \mathbb{E}\{\Pi_{\text{IN}r,t}\} &= \mathbb{E}\{\Pi_{\text{dr},t}\} + \mathbb{E}\{\Pi_{rs,t}\}.\end{aligned}$$

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Basic SEA equations

Average power flow

- It can be shown that:

$$\begin{aligned}\mathbb{E}\{\Pi_{rs,t}\} = & \sum_{\alpha=1}^{+\infty} \sum_{\beta=1}^{+\infty} \int_{I_0 \cup \underline{I}_0} a_{r,\alpha}^{s,\beta}(\omega) [E_{\alpha\alpha}^r(\omega) - E_{\beta\beta}^s(\omega)] d\omega \\ & + \sum_{\alpha=1}^{+\infty} \sum_{\beta=1}^{+\infty} \left(\sum_{\substack{\alpha' = 1 \\ \alpha' \neq \alpha}}^{+\infty} \int_{I_0 \cup \underline{I}_0} C_{r,\alpha\alpha'}^{s,\beta}(\omega) E_{\alpha\alpha'}^r(\omega) d\omega \right. \\ & \left. - \sum_{\substack{\beta' = 1 \\ \beta' \neq \beta}}^{+\infty} \int_{I_0 \cup \underline{I}_0} C_{s,\beta\beta'}^{r,\alpha}(\omega) E_{\beta\beta'}^s(\omega) d\omega \right)\end{aligned}$$

where:

$$E_{\alpha\alpha'}^r(\omega) = \Re\{\omega^2 M_r S_{\alpha\alpha'}^r \hat{h}_{r\alpha}(\omega) \hat{h}_{r\alpha'}^*(\omega)\},$$

$$S_{\alpha\alpha'}^r = \frac{1}{M_r} \int_{\Omega_r} \int_{\Omega_r} (\mathbf{S}_r(\mathbf{x}, \mathbf{y}) \phi_{r,\alpha'}(\mathbf{y}), \phi_{r,\alpha}(\mathbf{x})) d\mathbf{x} d\mathbf{y},$$

$$\hat{h}_{r\alpha}(\omega) = [M_r(\omega_{r\alpha}^2 - \omega^2 + i\eta_{r\alpha}\omega_{r\alpha}\omega)]^{-1}.$$

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- The formula can be simplified invoking two assumptions:

- 1 Either "rain-on-the-roof" excitations, $S_{\alpha\alpha'}^r = S_{r\alpha}\delta_{\alpha\alpha'}$ for some $S_{r\alpha} > 0$;
- 2 Or "weak coupling", s.t. $\mathcal{E}(t) \simeq \mathcal{E}_1(t) + \mathcal{E}_2(t)$ neglecting the contribution of the coupling operators to the mechanical energy of the overall system.

- Then:

$$\mathbb{E}\{\Pi_{rs,t}\} = \sum_{\alpha=1}^{+\infty} \sum_{\beta=1}^{+\infty} \int_{I_0 \cup \underline{I}_0} a_{r,\alpha}^{s,\beta}(\omega) [E_{\alpha\alpha}^r(\omega) - E_{\beta\beta}^s(\omega)] d\omega.$$

- The coefficients $a_{r,\alpha}^{s,\beta}(\omega) = a_{s,\beta}^{r,\alpha}(\omega)$ (and $C_{r,\alpha\alpha'}^{s,\beta}(\omega)$) are independent of the loads.

Basic SEA equations

Average power flow

- Let $\mathcal{J}_r = \{\alpha; \omega_{r\alpha} \in I_0\}$, for $r \in \{1, 2\}$.

- **Hypotheses** (wideband excitations):

- i** $\xi_{r\alpha} \ll 1$, $\forall \alpha \in \mathcal{J}_r$, $r \in \{1, 2\}$;

- ii** $\Delta\omega \gg b_{r\alpha} = \pi \xi_{r\alpha} \omega_{r\alpha}$, $\forall \alpha \in \mathcal{J}_r$, $r \in \{1, 2\}$;

- iii** The average mechanical energy of the stationary forced response for each sub-system is due to the modes in $\mathcal{J}_r$ solely, $r \in \{1, 2\}$.

- Then:

$$\mathbb{E}\{\Pi_{rs,t}\} \simeq \sum_{\alpha \in \mathcal{J}_r} \sum_{\beta \in \mathcal{J}_s} a_{r,\alpha}^{s,\beta}(\omega_0) \left[\frac{\pi S_{r\alpha}}{D_{r\alpha}} - \frac{\pi S_{s\beta}}{D_{s\beta}} \right].$$

- This is the analog of the result obtained for the two-DOFs system, summing the contributions of all modes of each sub-system within the frequency band of analysis I_0 .

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Internal fluid-structure interaction

Notations and setting

- The structure: we consider small perturbations $\mathbf{u}_1(\mathbf{x}, t)$ around a static equilibrium $\mathbf{x} \in \Omega_1$ considered as the reference configuration.
- The structure occupying Ω_1 is constituted by linear, memoryless viscoelastic materials:

$$\boldsymbol{\sigma}_1(\mathbf{x}, t) = \mathbf{C}_1^e \boldsymbol{\epsilon}_1(\mathbf{x}, t) + \mathbf{C}_1^v \dot{\boldsymbol{\epsilon}}_1(\mathbf{x}, t), \quad (\mathbf{x}, t) \in \Omega_1 \times \mathbb{R},$$

where $\boldsymbol{\epsilon}_1 = \nabla \otimes_s \mathbf{u}_1$ is the small strain tensor, $\boldsymbol{\sigma}_1$ the Cauchy stress tensor, $\mathbf{C}_1^e$ the elasticity tensor, $\mathbf{C}_1^v$ the viscosity tensor, and ϱ_1 the density.

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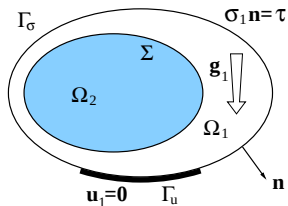
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Notations and setting



- The balance of momentum, boundary conditions and initial conditions read:

$$\left\{ \begin{array}{ll} \mathbf{Div} \boldsymbol{\sigma}_1 + \varrho_1 \mathbf{g}_1 = \varrho_1 \frac{\partial^2 \mathbf{u}_1}{\partial t^2} & \text{in } \Omega_1, \\ \mathbf{u}_1 = \mathbf{0} & \text{on } \Gamma_u, \\ \boldsymbol{\sigma}_1 \mathbf{n} = \boldsymbol{\tau}_1 & \text{on } \Gamma_\sigma, \\ \boldsymbol{\sigma}_1 \mathbf{n} = -p_2 \mathbf{n} & \text{on } \Sigma, \\ \mathbf{u}_1(\cdot, 0) = \mathbf{u}_0 & \text{in } \Omega_1, \\ \dot{\mathbf{u}}_1(\cdot, 0) = \mathbf{v}_0 & \text{in } \Omega_1, \end{array} \right.$$

where $\mathbf{n}$ is the unit outward normal to $\partial\Omega_1$.

Internal fluid-structure interaction

Notations and setting

- The acoustic fluid: homogeneous, compressible, inviscid, gravity effects are neglected. It occupies the bounded domain Ω_2 where $\partial\Omega_2 = \Sigma$.
- Irrotational motion $\nabla \times \mathbf{v}_2 = \mathbf{0}$, s.t. the fluid velocity $\mathbf{v}_2$ reads $\mathbf{v}_2 = \nabla\psi_2$ and the fluid pressure reads $p_2 = -\varrho_2\partial_t\psi_2 + \pi$.
- The **static correction**:

$$\pi = \frac{\varrho_2 c_2^2}{|\Omega_2|} \int_{\Sigma} \mathbf{u}_1 \cdot \mathbf{n} \, d\sigma,$$

where ϱ_2 : fluid density, c_2 : sound speed, accounts for the pressure variation induced by a static deformation of Σ .

- **Hypothesis**: π is negligible (*e.g.* air cavity).

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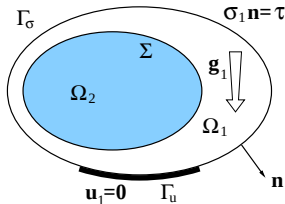
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Notations and setting



- The linearized Euler equation, boundary conditions and initial conditions yield for the **velocity potential** ψ_2 :

$$\left\{ \begin{array}{ll} \frac{1}{c_2^2} \frac{\partial^2 \psi_2}{\partial t^2} - \Delta \psi_2 = g_2 & \text{in } \Omega_2, \\ \frac{\partial \psi_2}{\partial \mathbf{n}} = \dot{\mathbf{u}}_1 \cdot \mathbf{n} & \text{on } \Sigma, \\ \psi_2(\cdot, 0) = 0 & \text{in } \Omega_2, \\ \dot{\psi}_2(\cdot, 0) = 0 & \text{in } \Omega_2. \end{array} \right.$$

Internal fluid-structure interaction

Virtual Power Principle

- The Virtual Power principle (VPP) on the sets of admissible fields

$$\mathcal{C}_1 = \{ \mathbf{v}; \mathbf{v} \in L^2(\Omega_1), \nabla \mathbf{v} \in L^2(\Omega_1), \text{ and } \mathbf{v}|_{\Gamma_u} = \mathbf{0} \},$$

$$\mathcal{C}_2 = \{ \phi; \phi \in L^2(\Omega_2), \nabla \phi \in L^2(\Omega_2), \text{ and } \int_{\Omega_2} \phi \, d\mathbf{x} = 0 \},$$

reads: Find $(\mathbf{u}_1, \psi_2) \in \mathcal{C}_1 \times \mathcal{C}_2$ s.t.

$$\begin{aligned} & \int_{\Omega_1} \varrho_1 \ddot{\mathbf{u}}_1 \cdot \mathbf{v} \, d\mathbf{x} + \int_{\Omega_1} \boldsymbol{\sigma}_1(\mathbf{u}_1) : \boldsymbol{\epsilon}(\mathbf{v}) \, d\mathbf{x} - \int_{\Sigma} \varrho_2 \dot{\psi}_2 (\mathbf{n} \cdot \mathbf{v}) \, d\sigma \\ &= \int_{\Omega_1} \varrho_1 \mathbf{g}_1 \cdot \mathbf{v} \, d\mathbf{x} + \int_{\Gamma_\sigma} \boldsymbol{\tau}_1 \cdot \mathbf{v} \, d\sigma, \quad \forall \mathbf{v} \in \mathcal{C}_1, \\ & \int_{\Omega_2} \frac{\varrho_2}{c_2^2} \ddot{\psi}_2 \phi \, d\mathbf{x} + \int_{\Omega_2} \varrho_2 \nabla \psi_2 \cdot \nabla \phi \, d\mathbf{x} + \int_{\Sigma} \varrho_2 (\dot{\mathbf{u}}_1 \cdot \mathbf{n}) \phi \, d\sigma \\ &= \int_{\Omega_2} \varrho_2 g_2 \phi \, d\mathbf{x}, \quad \forall \phi \in \mathcal{C}_2. \end{aligned}$$

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- The structural mass, stiffness and damping bilinear forms:

$$m_1(\mathbf{u}, \mathbf{v}) = \langle \mathbf{M}_1 \mathbf{u}, \mathbf{v} \rangle_{\mathcal{C}'_1, \mathcal{C}_1} = \int_{\Omega_1} \varrho_1 \mathbf{u} \cdot \mathbf{v} \, dx,$$

$$k_1(\mathbf{u}, \mathbf{v}) = \langle \mathbf{K}_1 \mathbf{u}, \mathbf{v} \rangle_{\mathcal{C}'_1, \mathcal{C}_1} = \int_{\Omega_1} \mathbf{C}_1^e(\boldsymbol{\epsilon}(\mathbf{u})) : \boldsymbol{\epsilon}(\mathbf{v}) \, dx,$$

$$d_1(\mathbf{u}, \mathbf{v}) = \langle \mathbf{D}_1 \mathbf{u}, \mathbf{v} \rangle_{\mathcal{C}'_1, \mathcal{C}_1} = \int_{\Omega_1} \mathbf{C}_1^v(\boldsymbol{\epsilon}(\mathbf{u})) : \boldsymbol{\epsilon}(\mathbf{v}) \, dx,$$

define the positive definite, symmetric mass $\mathbf{M}_1$, stiffness $\mathbf{K}_1$ and damping $\mathbf{D}_1$ operators of $\mathcal{L}(\mathcal{C}_1, \mathcal{C}'_1)$ (the set of continuous operators), where $\langle \mathbf{f}, \mathbf{v} \rangle_{\mathcal{C}'_1, \mathcal{C}_1}$ defines the duality product of $\mathbf{f} \in \mathcal{C}'_1$ and $\mathbf{v} \in \mathcal{C}_1$, and $\mathcal{C}'_1$ is the dual space of $\mathcal{C}_1$ (the set of linear forms).

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- The fluid mass and stiffness bilinear forms:

$$m_2(\psi, \phi) = \langle \mathbf{M}_2 \psi, \phi \rangle_{\mathcal{C}'_2, \mathcal{C}_2} = \int_{\Omega_2} \frac{\varrho_2}{c_2^2} \psi \phi \, d\mathbf{x},$$

$$k_2(\psi, \phi) = \langle \mathbf{K}_2 \psi, \phi \rangle_{\mathcal{C}'_2, \mathcal{C}_2} = \int_{\Omega_2} \varrho_2 \nabla \psi \cdot \nabla \phi \, d\mathbf{x},$$

define the positive definite, symmetric mass $\mathbf{M}_2$ and stiffness $\mathbf{K}_2$ operators of $\mathcal{L}(\mathcal{C}_2, \mathcal{C}'_2)$.

- The **gyroscopic coupling** linear form on $\mathcal{C} = \mathcal{C}_1 \times \mathcal{C}_2$:

$$d_c \left(\begin{array}{c} \mathbf{u} \\ \psi \end{array} \middle| \begin{array}{c} \mathbf{v} \\ \phi \end{array} \right) = \left\langle \begin{array}{c} \boldsymbol{\Omega} \\ \psi \end{array} \middle| \begin{array}{c} \mathbf{v} \\ \phi \end{array} \right\rangle_{\mathcal{C}', \mathcal{C}} = - \int_{\Sigma} \varrho_2 \psi (\mathbf{v} \cdot \mathbf{n}) \, d\sigma + \int_{\Sigma} \varrho_2 \phi (\mathbf{u} \cdot \mathbf{n}) \, d\sigma$$

is skew-symmetric (thus conservative).

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- By the Riesz theorem, $\exists! \mathbf{f}_1 \in \mathcal{C}'_1$ and $\exists! \mathbf{f}_2 \in \mathcal{C}'_2$ s.t.

$$\begin{aligned} f_1(\mathbf{v}) &= \langle \mathbf{f}_1, \mathbf{v} \rangle_{\mathcal{C}'_1, \mathcal{C}_1} = \int_{\Omega_1} \varrho_1 \mathbf{g}_1 \cdot \mathbf{v} \, d\mathbf{x} + \int_{\Gamma_\sigma} \boldsymbol{\tau}_1 \cdot \mathbf{v} \, d\sigma, & \forall \mathbf{v} \in \mathcal{C}_1, \\ f_2(\phi) &= \langle f_2, \phi \rangle_{\mathcal{C}'_2, \mathcal{C}_2} = \int_{\Omega_2} \varrho_2 g_2 \phi \, d\mathbf{x}, & \forall \phi \in \mathcal{C}_2. \end{aligned}$$

- Then the VPP reads:

$$\begin{bmatrix} \mathbf{M}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{M}_2 \end{bmatrix} \begin{pmatrix} \ddot{\mathbf{u}}_1 \\ \ddot{\psi}_2 \end{pmatrix} + \begin{bmatrix} \mathbf{D}_1 & \boldsymbol{\Omega}_{12} \\ \boldsymbol{\Omega}_{21} & \mathbf{0} \end{bmatrix} \begin{pmatrix} \dot{\mathbf{u}}_1 \\ \dot{\psi}_2 \end{pmatrix} + \begin{bmatrix} \mathbf{K}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{K}_2 \end{bmatrix} \begin{pmatrix} \mathbf{u}_1 \\ \psi_2 \end{pmatrix} = \begin{pmatrix} \mathbf{f}_1 \\ f_2 \end{pmatrix},$$

”in the sense of distributions”, with some prescribed initial conditions.

- This is the situation addressed beforehand, when the mass and stiffness coupling operators vanish:

$$\mathbf{M}_{12} = \mathbf{M}_{21} = \mathbf{0}, \quad \mathbf{K}_{12} = \mathbf{K}_{21} = \mathbf{0}.$$

Internal fluid-structure interaction

Eigenmodes and modal expansion

- **Spectral problem** for the structure: Find $\lambda \in \mathbb{R}$ and $\phi \in \mathcal{C}_1$ s.t.

$$\mathbf{K}_1 \phi = \lambda \mathbf{M}_1 \phi.$$

- It admits a countable set of solutions $(\lambda_{1,1}, \phi_{1,1})$, $(\lambda_{1,2}, \phi_{1,2}) \dots$ s.t. $0 < \lambda_{1,1} \leq \lambda_{1,2} \leq \dots$ and $\{\phi_{1\alpha}\}_{\alpha \in \mathbb{N}^*}$ is an Hilbertian basis of the space $H_1 = L^2_\mu(\Omega_1)$ of square integrable functions with respect to the unit mass measure $\mu_1(d\mathbf{x}) = \mathbb{1}_{\Omega_1} \frac{\varrho_1 d\mathbf{x}}{M_1}$, with $M_1 = \int_{\Omega_1} \varrho_1 d\mathbf{x}$:

$$m_1(\phi_{1\alpha}, \phi_{1\alpha'}) = M_1 \delta_{\alpha\alpha'},$$

$$k_1(\phi_{1\alpha}, \phi_{1\alpha'}) = M_1 \lambda_{1\alpha} \delta_{\alpha\alpha'}.$$

- $\{\phi_{1\alpha}\}_{\alpha \in \mathbb{N}^*}$ are the **structural modes *in vacuo***, which are assumed (Basile) to diagonalize the damping operator $\mathbf{D}_1$ as well.

Internal fluid-structure interaction

Eigenmodes and modal expansion

- **Spectral problem** for the acoustic cavity: Find $\lambda \in \mathbb{R}$ and $\phi \in \mathcal{C}_2$ s.t.

$$\mathbf{K}_2 \phi = \lambda \mathbf{M}_2 \phi.$$

- It admits a countable set of solutions $(\lambda_{2,1}, \phi_{2,1})$, $(\lambda_{2,2}, \phi_{2,2}) \dots$ s.t. $0 < \lambda_{2,1} \leq \lambda_{2,2} \leq \dots$ and $\{\phi_{2\beta}\}_{\alpha \in \mathbb{N}^*}$ is an Hilbertian basis of the space $H_2 = L^2(\Omega_2)$ of square integrable functions with respect to the scalar product $\mu_2(\psi, \phi) = \int_{\Omega_2} \psi \phi \, d\mathbf{x}$, with $M_2 = \int_{\Omega_2} \varrho_2 \, d\mathbf{x}$:

$$m_2(\phi_{2\beta}, \phi_{2\beta'}) = \frac{\varrho_2}{c_2^2} \mu_2(\phi_{2\beta}, \phi_{2\beta'}) = \frac{M_2}{c_2^2} \delta_{\beta\beta'},$$

$$k_2(\phi_{2\beta}, \phi_{2\beta'}) = \frac{M_2}{c_2^2} \lambda_{2\beta} \delta_{\beta\beta'}.$$

- $\{\phi_{2\beta}\}_{\beta \in \mathbb{N}^*}$ are the **rigid-wall acoustic modes**.

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- Consequently, the solution $\mathbf{u}_1 \in \mathcal{C}_1$ can be expanded on the eigenbasis $\{\phi_{1\alpha}\}_{\alpha \in \mathbb{N}^*}$ as:

$$\mathbf{u}_1(\mathbf{x}, t) = \sum_{\alpha=1}^{\infty} q_{1\alpha}(t) \phi_{1\alpha}(\mathbf{x}), \quad (\mathbf{x}, t) \in \Omega_1 \times \mathbb{R},$$

- Likewise, the solution $\psi_2 \in \mathcal{C}_2$ can be expanded on the eigenbasis $\{\phi_{2\beta}\}_{\beta \in \mathbb{N}^*}$ as:

$$\psi_2(\mathbf{x}, t) = \sum_{\beta=1}^{\infty} q_{2\beta}(t) \phi_{2\beta}(\mathbf{x}), \quad (\mathbf{x}, t) \in \Omega_2 \times \mathbb{R}.$$

Internal fluid-structure interaction

Eigenmodes and modal expansion

- The associated mechanical energies for both sub-systems are:

$$\mathcal{E}_1(t) = \frac{M_1}{2} \sum_{\alpha=1}^{+\infty} [(\dot{q}_{1\alpha}(t))^2 + \omega_{1\alpha}^2 (q_{1\alpha}(t))^2] ,$$

$$\mathcal{E}_2(t) = \frac{M_2}{2c_2^2} \sum_{\alpha=1}^{+\infty} [(\dot{q}_{2\alpha}(t))^2 + \omega_{2\alpha}^2 (q_{2\alpha}(t))^2] ,$$

where $\omega_{r\alpha} = \sqrt{\lambda_{r\alpha}}$ is the "blocked" (natural) frequency of the α^{th} mode of the r^{th} sub-system—*i.e.* the α^{th} structural mode *in vacuo* for $r = 1$ or the α^{th} rigid-wall acoustic mode for $r = 2$.

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- The average mechanical energies of the forced responses under mean-square stationary random loads are:

$$\mathbb{E}\{\mathcal{E}_{1,t}\} = M_1 \int_{\mathbb{R}} \omega^2 \text{Tr} \mathbf{S}_{U_1}(\omega) d\omega ,$$
$$\mathbb{E}\{\mathcal{E}_{2,t}\} = \frac{M_2}{c_2^2} \int_{\mathbb{R}} S_{P_2}(\omega) d\omega ,$$

where $\omega \mapsto \mathbf{S}_{U_1}(\omega)$ is the spectral density matrix of the structure displacement, and $\omega \mapsto S_{P_2}(\omega)$ is the spectral density function of the fluid pressure.

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- The structure: we consider small perturbations $\mathbf{u}_s(\mathbf{x}, t)$ around a static equilibrium $\mathbf{x} \in \Omega_s$ considered as the reference configuration.
- The structure occupying Ω_s is constituted by linear, memoryless viscoelastic materials:

$$\boldsymbol{\sigma}_s(\mathbf{x}, t) = \mathbf{C}_s^e \boldsymbol{\epsilon}_s(\mathbf{x}, t) + \mathbf{C}_s^v \dot{\boldsymbol{\epsilon}}_s(\mathbf{x}, t), \quad (\mathbf{x}, t) \in \Omega_s \times \mathbb{R},$$

where $\boldsymbol{\epsilon}_s = \nabla \otimes_s \mathbf{u}_s$ is the small strain tensor, $\boldsymbol{\sigma}_s$ the Cauchy stress tensor, $\mathbf{C}_s^e$ the elasticity tensor, $\mathbf{C}_s^v$ the viscosity tensor, and ρ_s the density.

Notations and setting



- $$\left\{ \begin{array}{ll} \mathbf{Div} \boldsymbol{\sigma}_s + \varrho_s \mathbf{g}_s = \varrho_s \frac{\partial^2 \mathbf{u}_s}{\partial t^2} & \text{in } \Omega_s, \\ \mathbf{u}_s = \mathbf{0} & \text{on } \Gamma_u, \\ \boldsymbol{\sigma}_s \mathbf{n} = \boldsymbol{\tau}_s & \text{on } \Gamma_\sigma, \\ \boldsymbol{\sigma}_s \mathbf{n} = -p_f \mathbf{n} & \text{on } \Gamma, \\ \mathbf{u}_s(\cdot, 0) = \mathbf{u}_0 & \text{in } \Omega_s, \\ \dot{\mathbf{u}}_s(\cdot, 0) = \mathbf{v}_0 & \text{in } \Omega_s, \end{array} \right.$$

final to $\mathcal{O}M_s$.

External fluid-structure interaction

Notations and setting

- The acoustic fluid: homogeneous, compressible, inviscid, gravity effects are neglected.
- Irrotational motion $\nabla \times \mathbf{v}_f = \mathbf{0}$, s.t. the fluid velocity $\mathbf{v}_f$ reads $\mathbf{v}_f = \nabla \psi_f$ and the fluid pressure reads $p_f = -\varrho_f \partial_t \psi_f$, where the **velocity potential** ψ_f satisfies:

$$\left\{ \begin{array}{ll} \frac{1}{c_f^2} \frac{\partial^2 \psi_f}{\partial t^2} - \Delta \psi_f = g_{IN} & \text{in } \Omega_f, \\ \frac{\partial \psi_f}{\partial \mathbf{n}} = \dot{\mathbf{u}}_s \cdot \mathbf{n} & \text{on } \Gamma, \\ \psi_f(\cdot, 0) = 0 & \text{in } \Omega_f, \\ \dot{\psi}_f(\cdot, 0) = 0 & \text{in } \Omega_f, \end{array} \right.$$

with ϱ_f : fluid density, c_f : sound speed.

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Decomposition of the velocity potential

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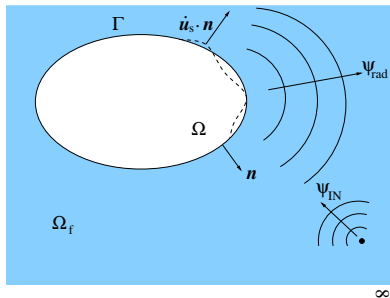
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- The velocity potential reads:

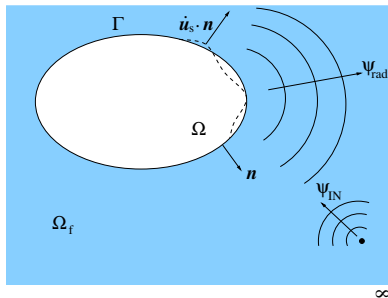
$$\psi_f(\mathbf{x}, t) = \psi_{IN}(\mathbf{x}, t) + \psi_{d0}(\mathbf{x}, t) + \psi_{rad}(\mathbf{x}, t), \quad (\mathbf{x}, t) \in \Omega_f \times \mathbb{R},$$

where:

- ψ_{IN} : incident potential (data);
- ψ_{d0} : diffracted potential, the structure being motionless;
- ψ_{rad} : radiated potential.

External fluid-structure interaction

Decomposition of the velocity potential

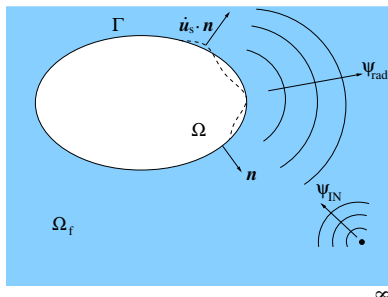


- The diffracted potential ψ_{d0} satisfies:

$$\left\{ \begin{array}{ll} \frac{1}{c_f^2} \frac{\partial^2 \psi_{d0}}{\partial t^2} - \Delta \psi_{d0} = 0 & \text{in } \Omega_f, \\ \frac{\partial \psi_{d0}}{\partial n} = -\frac{\partial \psi_{IN}}{\partial n} & \text{on } \Gamma. \end{array} \right.$$

External fluid-structure interaction

Decomposition of the velocity potential



- The radiated potential ψ_{rad} satisfies:

$$\left\{ \begin{array}{l} \frac{1}{c_f^2} \frac{\partial^2 \psi_{\text{rad}}}{\partial t^2} - \Delta \psi_{\text{rad}} = 0 \quad \text{in } \Omega_f, \\ \frac{\partial \psi_{\text{rad}}}{\partial n} = \dot{\mathbf{u}}_s \cdot \mathbf{n} \quad \text{on } \Gamma. \end{array} \right.$$

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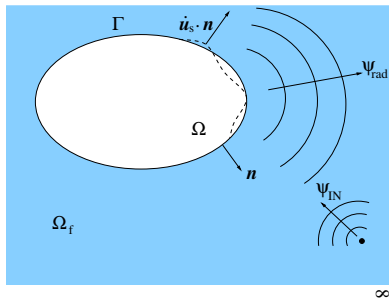
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- The **incident potential** ψ_{IN} satisfies:

$$\frac{1}{c_f^2} \frac{\partial^2 \psi_{\text{IN}}}{\partial t^2} - \Delta \psi_{\text{IN}} = g_{\text{IN}}(\mathbf{x}, t), \quad (\mathbf{x}, t) \in \mathbb{R}^3 \times \mathbb{R},$$

for some sound source g_{IN} in the full physical space.

Solving the fluid equations

Exterior Helmholtz problem

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- The exterior Helmholtz problem:

$$\left\{ \begin{array}{ll} \Delta\psi + k_f^2\psi = 0 & \text{in } \Omega_f, \\ \frac{\partial\psi}{\partial\mathbf{n}} = v & \text{on } \partial\Omega_f, \\ |\psi| = O\left(\frac{1}{r}\right), \quad \left|\frac{\partial\psi}{\partial r} + ik_f\psi\right| = O\left(\frac{1}{r^2}\right) & \text{as } r = \|\mathbf{x}\| \rightarrow +\infty, \end{array} \right.$$

where $k_f = \frac{\omega}{c_f}$, and $\partial\Omega_f = \Gamma$ is actually the interface between the fluid and the structure.

- Sommerfeld radiation conditions "at infinity": the radiated waves are almost plane and do not propagate toward Γ .

Solving the fluid equations

Boundary impedance

- The exterior Helmholtz problem admits a unique solution $\forall k_f \in \mathbb{R}$, hence:

- There exists a linear operator $\mathcal{B}_\Gamma(k_f) : \mathcal{C}'_\Gamma \rightarrow \mathcal{C}_\Gamma$ s.t.:

$$\psi|_\Gamma(\omega) = \mathcal{B}_\Gamma(k_f)v, \quad \text{on } \Gamma;$$

- There exists a linear operator $\mathcal{R}_x(k_f) : \mathcal{C}'_\Gamma \rightarrow \mathbb{C}$ s.t.:

$$\psi(x, \omega) = \mathcal{R}_x(k_f)v, \quad x \in \Omega_f,$$

with $\mathcal{C}_\Gamma$: the set of admissible fields on Γ ($\mathcal{C}'_\Gamma$: its dual).

- $\mathcal{Z}_\Gamma(\omega) = -i\omega\rho_f\mathcal{B}_\Gamma(k_f)$ is the **acoustic impedance boundary operator**.
- $\mathcal{Z}_x(\omega) = -i\omega\rho_f\mathcal{R}_x(k_f)$ is the **radiation impedance operator**.

Solving the fluid equations

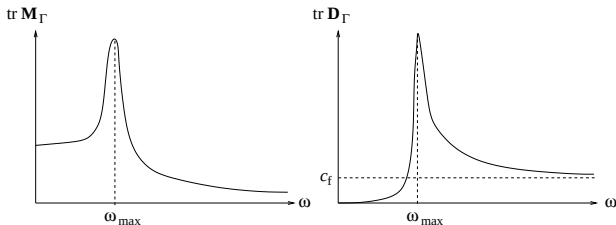
Boundary impedance

- The boundary impedance $\mathbf{Z}_\Gamma(\omega)$ is symmetric and reads:

$$i\omega \mathbf{Z}_\Gamma(\omega) = -\omega^2 \mathbf{M}_\Gamma \left(\frac{\omega}{c_f} \right) + i\omega \mathbf{D}_\Gamma \left(\frac{\omega}{c_f} \right),$$

where:

- The **reactive part** $\omega \mapsto \text{Tr } \mathbf{M}_\Gamma \left(\frac{\omega}{c_f} \right)$ (left) is generally unsigned, though it is positive if $\mathbb{R}^3 \setminus \overline{\Omega}_f$ is convex;
- The **resistive part** $\omega \mapsto \text{Tr } \mathbf{D}_\Gamma \left(\frac{\omega}{c_f} \right)$ (right) is positive.



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- Diffracted velocity potential:

$$\begin{aligned}\hat{\psi}_{\text{d0}}(\omega)|_{\Gamma} &= -\mathcal{B}_{\Gamma}(k_f) \frac{\partial \hat{\psi}_{\text{IN}}}{\partial \mathbf{n}} \quad \mathbf{x} \in \Gamma, \\ \hat{\psi}_{\text{d0}}(\mathbf{x}, \omega) &= -\mathcal{R}_{\mathbf{x}}(\omega) \frac{\partial \hat{\psi}_{\text{IN}}}{\partial \mathbf{n}} \quad \mathbf{x} \in \Omega_f.\end{aligned}$$

- Radiated velocity potential:

$$\begin{aligned}\hat{\psi}_{\text{rad}}(\omega)|_{\Gamma} &= i\omega \mathcal{B}_{\Gamma}(k_f) (\hat{\mathbf{u}}_s(\omega) \cdot \mathbf{n}) \quad \mathbf{x} \in \Gamma, \\ \hat{\psi}_{\text{rad}}(\mathbf{x}, \omega) &= i\omega \mathcal{R}_{\mathbf{x}}(\omega) (\hat{\mathbf{u}}_s(\omega) \cdot \mathbf{n}) \quad \mathbf{x} \in \Omega_f.\end{aligned}$$

- The (linear) scattering operator $\mathcal{T}_{\Gamma}(\omega) \equiv \mathbf{I} - \mathcal{B}_{\Gamma}(k_f) \partial_{\mathbf{n}}$ s.t.:

$$(\hat{\psi}_{\text{IN}}(\omega) + \hat{\psi}_{\text{d0}}(\omega))|_{\Gamma} = \mathcal{T}_{\Gamma}(\omega) \hat{\psi}_{\text{IN}}(\omega)|_{\Gamma}.$$

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- The Virtual Power Principle (VPP) on the set of admissible displacement fields:

$$\mathcal{C}_s = \{ \mathbf{v}; \mathbf{v} \in L^2(\Omega_s), \nabla \mathbf{v} \in L^2(\Omega_s), \text{ and } \mathbf{v}|_{\Gamma_u} = \mathbf{0} \},$$

reads for the structure, in the frequency domain: Find $\hat{\mathbf{u}}_s \in \mathcal{C}_s$ s.t.

$$\begin{aligned} & -\omega^2 \int_{\Omega_s} \varrho_s \hat{\mathbf{u}}_s \cdot \bar{\mathbf{v}} \, d\mathbf{x} + i\omega \int_{\Omega_s} \mathbf{C}_s^v \boldsymbol{\epsilon}(\hat{\mathbf{u}}_s) : \boldsymbol{\epsilon}(\bar{\mathbf{v}}) \, d\mathbf{x} + \int_{\Omega_s} \mathbf{C}_s^e \boldsymbol{\epsilon}(\hat{\mathbf{u}}_s) : \boldsymbol{\epsilon}(\bar{\mathbf{v}}) \, d\mathbf{x} \\ & - i\omega \int_{\Gamma} \varrho_f \hat{\psi}_f(\bar{\mathbf{v}} \cdot \mathbf{n}) \, d\sigma = \int_{\Omega_s} \varrho_s \hat{\mathbf{g}}_s \cdot \bar{\mathbf{v}} \, d\mathbf{x} + \int_{\Gamma_\sigma} \hat{\boldsymbol{\tau}}_s \cdot \bar{\mathbf{v}} \, d\sigma, \quad \forall \mathbf{v} \in \mathcal{C}_s. \end{aligned}$$

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- The structural mass, stiffness and damping bilinear forms:

$$m_s(\mathbf{u}, \mathbf{v}) = \langle \mathbf{M}_s \mathbf{u}, \mathbf{v} \rangle_{\mathcal{C}'_s, \mathcal{C}_s} = \int_{\Omega_s} \rho_s \mathbf{u} \cdot \bar{\mathbf{v}} \, dx,$$

$$k_s(\mathbf{u}, \mathbf{v}) = \langle \mathbf{K}_s \mathbf{u}, \mathbf{v} \rangle_{\mathcal{C}'_s, \mathcal{C}_s} = \int_{\Omega_s} \mathbf{C}_s^e(\boldsymbol{\epsilon}(\mathbf{u})) : \boldsymbol{\epsilon}(\bar{\mathbf{v}}) \, dx,$$

$$d_s(\mathbf{u}, \mathbf{v}) = \langle \mathbf{D}_s \mathbf{u}, \mathbf{v} \rangle_{\mathcal{C}'_s, \mathcal{C}_s} = \int_{\Omega_s} \mathbf{C}_s^v(\boldsymbol{\epsilon}(\mathbf{u})) : \boldsymbol{\epsilon}(\bar{\mathbf{v}}) \, dx,$$

- The boundary coupling bilinear form:

$$b_\Gamma(\mathbf{u}, \mathbf{v}; \omega) = \langle \mathbf{Z}_\Gamma(\omega) \mathbf{u}, \mathbf{v} \rangle_{\mathcal{C}'_s, \mathcal{C}_s} = \int_\Gamma (\mathbf{Z}_\Gamma(\omega) \mathbf{u} \cdot \mathbf{n}, \bar{\mathbf{v}} \cdot \mathbf{n}) \, d\sigma.$$

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- By the Riesz theorem, $\exists! \hat{\mathbf{f}}_s, \hat{\mathbf{f}}_{\text{IN}} \in \mathcal{C}'_s$ s.t.

$$f_s(\mathbf{v}) = \langle \hat{\mathbf{f}}_s, \mathbf{v} \rangle_{\mathcal{C}'_s, \mathcal{C}_s} = \int_{\Omega_s} \varrho_s \hat{\mathbf{g}}_s \cdot \bar{\mathbf{v}} \, d\mathbf{x} + \int_{\Gamma_\sigma} \hat{\boldsymbol{\tau}}_s \cdot \bar{\mathbf{v}} \, d\sigma,$$

$$f_{\text{IN}}(\mathbf{v}) = \langle \hat{\mathbf{f}}_{\text{IN}}, \mathbf{v} \rangle_{\mathcal{C}'_s, \mathcal{C}_s} = - \int_{\Gamma} \mathcal{T}_\Gamma(\omega) \hat{p}_{\text{IN}}(\bar{\mathbf{v}} \cdot \mathbf{n}) \, d\sigma, \quad \forall \mathbf{v} \in \mathcal{C}_s.$$

- Then the VPP in the frequency domain reads:

$$\begin{aligned} \left[-\omega^2 \left(\mathbf{M}_s + \mathbf{M}_\Gamma \left(\frac{\omega}{c_f} \right) \right) + i\omega \left(\mathbf{D}_s + \mathbf{D}_\Gamma \left(\frac{\omega}{c_f} \right) \right) + \mathbf{K}_s \right] \hat{\mathbf{u}}_s(\omega) \\ = \hat{\mathbf{f}}_s(\omega) + \hat{\mathbf{f}}_e(\omega) + \hat{\mathbf{f}}_{\text{IN}}(\omega), \end{aligned}$$

”in the sense of distributions”.

- $\hat{\mathbf{f}}_e(\omega) = \mathbf{D}_s \mathbf{u}_0 + \mathbf{M}_s(i\omega \mathbf{u}_0 + \mathbf{v}_0)$ is an equivalent load accounting for the initial conditions.

External fluid-structure interaction

Eigenmodes and modal expansion

- **Spectral problem:** Find $\lambda \in \mathbb{R}$ and $\phi \in \mathcal{C}_s$ s.t.

$$K_s \phi = \lambda M_s \phi.$$

- It admits a countable set of solutions $(\lambda_{s1}, \phi_{s1})$, $(\lambda_{s2}, \phi_{s2}) \dots$ s.t. $0 < \lambda_{s1} \leq \lambda_{s2} \leq \dots$ and $\{\phi_{s\alpha}\}_{\alpha \in \mathbb{N}^*}$ is an Hilbertian basis of the space $H_s = L^2_\mu(\Omega_s)$ of square integrable functions with respect to the unit mass measure $\mu_s(d\mathbf{x}) = \mathbb{1}_{\Omega_s} \frac{\varrho_s d\mathbf{x}}{M_s}$, with $M_s = \int_{\Omega_s} \varrho_s d\mathbf{x}$:

$$m_s(\phi_{s\alpha}, \phi_{s\beta}) = M_s \delta_{\alpha\beta},$$

$$k_s(\phi_{s\alpha}, \phi_{s\beta}) = M_s \omega_{s\alpha}^2 \delta_{\alpha\beta},$$

where $\omega_{s\alpha}^2 = \lambda_{s\alpha}$.

- $\{\phi_{s\alpha}\}_{\alpha \in \mathbb{N}^*}$ are the **structural modes *in vacuo***.

External fluid-structure interaction

Eigenmodes and modal expansion

- Consequently, the solution $\mathbf{u}_s \in \mathcal{C}_s$ can be expanded on the eigenbasis $\{\phi_{s\alpha}\}_{\alpha \in \mathbb{N}^*}$ as:

$$\mathbf{u}_s(\mathbf{x}, t) = \sum_{\alpha=1}^{\infty} q_{\alpha}(t) \phi_{s\alpha}(\mathbf{x}).$$

- Introducing $\mu_s(\mathbf{u}, \mathbf{v}) = \int_{\Omega_s} \mathbf{u} \cdot \mathbf{v} \mu_s(\mathrm{d}\mathbf{x})$ (the scalar product in H_s), the generalized coordinates $\{q_{\alpha}\}_{\alpha \in \mathbb{N}^*}$ are:

$$q_{\alpha} = \mu_s(\mathbf{u}_s, \phi_{s\alpha}).$$

- The μ -norm $\|\mathbf{u}_s\|_{\mu} = \sqrt{\mu_s(\mathbf{u}_s, \mathbf{u}_s)}$ is obtained as:

$$\|\mathbf{u}_s(\cdot, t)\|_{\mu} = \left(\sum_{\alpha=1}^{+\infty} (q_{\alpha}(t))^2 \right)^{\frac{1}{2}} < +\infty.$$

External fluid-structure interaction

Eigenmodes and modal expansion

- **Hypothesis** (Basile): the eigenmodes $\{\phi_{s\alpha}\}_{\alpha \in \mathbb{N}^*}$ diagonalize the damping operator as well:

$$d_s(\phi_{s\alpha}, \phi_{s\beta}) = M_s \eta_{s\alpha} \omega_{s\alpha} \delta_{\alpha\beta},$$

where:

- $\omega_{s\alpha}$: the (angular) eigenfrequency of the α^{th} mode *in vacuo*,
- $\xi_{s\alpha}$: the modal critical damping rate of the α^{th} mode *in vacuo*,
- $\eta_{s\alpha} = 2\xi_{s\alpha}$: the modal loss factor of the α^{th} mode *in vacuo*.

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$$M_s(-\omega^2 + i\eta_{s\alpha}\omega_{s\alpha}\omega + \omega_{s\alpha}^2)\hat{q}_\alpha(\omega) = \hat{f}_\alpha(\omega) - \sum_{\beta=1}^{+\infty} (-\omega^2 M_{\alpha\beta}(\omega) + i\omega D_{\alpha\beta}(\omega))\hat{q}_\beta(\omega)$$
$$M_{\alpha\beta}(\omega) = \left\langle \mathbf{M}_{\Gamma} \left(\frac{\omega}{c_f} \right) \phi_{s\alpha}, \phi_{s\beta} \right\rangle_{\mathcal{C}'_s, \mathcal{C}_s},$$

$$D_{\alpha\beta}(\omega) = \left\langle D_{\Gamma} \left(\frac{\omega}{c_f} \right) \phi_{s\alpha}, \phi_{s\beta} \right\rangle_{C'_s, C_s}.$$

- The generalized coordinates get coupled by the boundary impedance operator.

External fluid-structure interaction

Wet eigenfrequencies

- The frequency response function of the structure coupled to the fluid reads:

$$\hat{\mathbf{h}}_{\text{tot}}(\omega) = \left[-\omega^2 \left(\mathbf{M}_s + \mathbf{M}_\Gamma \left(\frac{\omega}{c_f} \right) \right) + i\omega \left(\mathbf{D}_s + \mathbf{D}_\Gamma \left(\frac{\omega}{c_f} \right) \right) + \mathbf{K}_s \right]^{-1}.$$

- The "wet" eigenfrequencies of the structure coupled to the fluid are the solutions of:

$$\det \left[-\omega^2 \left(\mathbf{M}_s + \mathbf{M}_\Gamma \left(\frac{\omega}{c_f} \right) \right) + \mathbf{K}_s \right] = 0.$$

- This eigenvalue problem is often approached by:

$$\left[1 + \frac{M_{\alpha\alpha}(\omega)}{M_s} \right] \omega^2 = \omega_{s\alpha}^2,$$

neglecting the off-diagonal terms.

Stationary incident pressure

Forced response of the structure

- We consider $\mathbf{f}_e \equiv \mathbf{0}$ and $\mathbf{f}_s \equiv \mathbf{0}$, focusing on the structure forced response to a random incident pressure field.
- **Data:** $(P_t, t \in \mathbb{R})$ is a second order, centered stochastic process defined on $(\Omega, \mathcal{F}, P)$, indexed on $\mathbb{R}$, with values in $L^2(\Gamma)$, and mean-square (m.s.) stationary.
- **Hypothesis:** It is characterized by its **cross spectral density function** $\omega \mapsto S_P(\omega; \mathbf{x}, \mathbf{y}) : \mathbb{R} \rightarrow \mathbb{R}_+$, $\mathbf{x}, \mathbf{y} \in \Gamma$, which is positive, even, integrable on $\mathbb{R}_\omega$, and s.t.:

$$S_P(\omega; \mathbf{x}, \mathbf{y}) = S_P(\omega; \mathbf{y}, \mathbf{x})$$

$$S_P(\omega; \mathbf{x}, \mathbf{y}) = S_{\text{IN}}(\mathbf{x}, \mathbf{y}) \otimes \mathbb{1}_{I_0 \cup I_0}(\omega),$$

where:

$$I_0 \cup I_0 = \left[\omega_0 - \frac{\Delta\omega}{2}, \omega_0 + \frac{\Delta\omega}{2} \right] \cup \left[-\omega_0 - \frac{\Delta\omega}{2}, -\omega_0 + \frac{\Delta\omega}{2} \right].$$

Stationary incident pressure

Forced response of the structure

- The forced response of the structure $t \mapsto \mathbf{u}_s^f(\cdot, t)$ is modeled by a stochastic process $(\mathbf{U}_t^f, t \in \mathbb{R})$ the properties of which are derived from filtering and mean-square derivation (see Lecture #1 part A).

Proposition

- $(\mathbf{U}_t^f, t \in \mathbb{R})$ is a second order, centered stochastic process defined on $(\Omega, \mathcal{F}, P)$, indexed on $\mathbb{R}$, with values in $\mathcal{C}_s$, and mean-square stationary s.t.:

$$\mathbf{S}_U(\omega; \mathbf{x}, \mathbf{y}) = \hat{\mathbf{h}}_{\text{tot}}(\omega) \mathcal{T}_{\mathbf{x}}(\omega) S_P(\omega; \mathbf{x}, \mathbf{y}) \mathcal{T}_{\mathbf{y}}(\omega)^* \hat{\mathbf{h}}_{\text{tot}}(\omega)^*.$$

- The same holds for its mean-square derivatives $(\dot{\mathbf{U}}_t^f, t \in \mathbb{R})$ and $(\ddot{\mathbf{U}}_t^f, t \in \mathbb{R})$, with values in H_s and $\mathcal{C}'_s$ respectively, s.t. $\mathbf{S}_{\dot{U}}(\omega) = \omega^2 \mathbf{S}_U(\omega)$, $\mathbf{S}_{\ddot{U}}(\omega) = \omega^4 \mathbf{S}_U(\omega)$.

Stationary incident pressure

Forced response of the structure

- Likewise, the modal forced responses $t \mapsto q_\alpha^f(t)$ for $\alpha \in \mathbb{N}^*$ are modeled by stochastic processes $(Q_{\alpha,t}^f, t \in \mathbb{R})$ the properties of which are derived from filtering and mean-square derivation (see Lecture #1 part A).

Proposition

- $(Q_{\alpha,t}^f, t \in \mathbb{R})$ is a $\mathbb{R}$ -valued second order, centered stochastic process defined on $(\Omega, \mathcal{F}, P)$, indexed on $\mathbb{R}$, and mean-square stationary.
- The same holds for its mean-square derivatives $(\dot{Q}_{\alpha,t}^f, t \in \mathbb{R})$ and $(\ddot{Q}_{\alpha,t}^f, t \in \mathbb{R})$.

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Forced response of the structure

- The finite-dimensional $\mathbb{C}^{N \times N}$ frequency response function $\hat{\mathbf{h}}_{\text{tot}}^N(\omega)$ of the structure coupled to the fluid:

$$[\hat{\mathbf{h}}_{\text{tot}}^N(\omega)]^{-1} = [\hat{\mathbf{h}}_s^N(\omega)]^{-1} + [\hat{\mathbf{h}}_\Gamma^N(\omega)]^{-1},$$

where for $0 < \alpha, \beta \leq N$:

$$[\hat{\mathbf{h}}_s^N(\omega)]_{\alpha\beta}^{-1} = M_s(\omega_{s\alpha}^2 - \omega^2 + i\eta_{s\alpha}\omega_{s\alpha}\omega)\delta_{\alpha\beta},$$

$$[\hat{\mathbf{h}}_\Gamma^N(\omega)]_{\alpha\beta}^{-1} = -\omega^2 M_{\alpha\beta}(\omega) + i\omega D_{\alpha\beta}(\omega).$$

- The $\mathbb{C}^{N \times N}$ power spectral density matrix $\mathbf{S}_Q$ of the $\mathbb{R}^N$ -valued stochastic process $\mathbf{Q}_t^f = (Q_{1,t}^f, \dots, Q_{N,t}^f)^\top$ thus reads:

$$\mathbf{S}_Q(\omega) = \hat{\mathbf{h}}_{\text{tot}}^N(\omega) \mathbf{S}_{\text{IN}}(\omega) \hat{\mathbf{h}}_{\text{tot}}^N(\omega)^* \otimes \mathbb{1}_{I_0 \cup I_0}(\omega),$$

with the $\mathbb{R}^{N \times N}$ symmetric real matrix $\mathbf{S}_{\text{IN}}$:

$$[\mathbf{S}_{\text{IN}}(\omega)]_{\alpha\beta} = \int_\Gamma \int_\Gamma (S_{\text{IN}}(\mathbf{x}, \mathbf{y}) \mathcal{T}_{\mathbf{y}}(\omega) \phi_\beta(\mathbf{y}), \mathcal{T}_{\mathbf{x}}(\omega) \phi_\alpha(\mathbf{x})) \, d\mathbf{x} d\mathbf{y}.$$

Stationary incident pressure

Energetics of the stationary forced response

- The average mechanical energy of the structure:

$$\begin{aligned}\mathbb{E}\{\mathcal{E}_t\} &= \frac{1}{2} \int_{I_0 \cup \underline{I}_0} \omega^2 \operatorname{Tr} \left[\left(\mathbf{M}_s + \mathbf{M}_\Gamma \left(\frac{\omega}{c_f} \right) \right) \mathbf{S}_Q(\omega) \right] d\omega \\ &\quad + \frac{1}{2} \int_{I_0 \cup \underline{I}_0} \operatorname{Tr} [\mathbf{K}_s \mathbf{S}_Q(\omega)] d\omega \\ &\stackrel{\text{def}}{=} |\Omega_s| \int_{I_0 \cup \underline{I}_0} \omega^2 (\varrho_s + \varrho_{\text{rad}}(\omega_0)) \operatorname{Tr} \mathbf{S}_Q(\omega) d\omega ,\end{aligned}$$

where $\varrho_{\text{rad}}(\omega_0)$ is an equivalent added density.

- The average **radiated power**:

$$\begin{aligned}\mathbb{E}\{\Pi_{\text{rad},t}\} &= \int_{I_0 \cup \underline{I}_0} \omega^2 \operatorname{Tr} \left[\mathbf{D}_\Gamma \left(\frac{\omega}{c_f} \right) \mathbf{S}_Q(\omega) \right] d\omega \\ &\stackrel{\text{def}}{=} \omega_0 \eta_{\text{rad}}(\omega_0) \mathbb{E}\{\mathcal{E}_t\} ,\end{aligned}$$

where $\eta_{\text{rad}}(\omega_0)$ is an equivalent added loss factor.

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- The average dissipated power thus reads:

$$\begin{aligned}\mathbb{E}\{\Pi_{d,t}\} &= \int_{I_0 \cup \underline{I}_0} \omega^2 \operatorname{Tr} [\mathbf{D}_s \mathbf{S}_Q(\omega)] d\omega \\ &= \frac{M_s}{M_s + \varrho_{\text{rad}}(\omega_0) |\Omega_s|} \omega_0 \eta_s(\omega_0) \mathbb{E}\{\mathcal{E}_t\},\end{aligned}$$

where $\eta_s(\omega_0)$ is the **average structural loss factor** in I_0 .

- The average input power:

$$\mathbb{E}\{\Pi_{\text{IN},t}\} = \Re \int_{I_0 \cup \underline{I}_0} i\omega \operatorname{Tr} \left[\hat{\mathbf{h}}_{\text{tot}}^N(\omega) \mathbf{S}_{\text{IN}}(\omega) \right] d\omega.$$

Stationary incident pressure

Power balance for the stationary forced response

- The instantaneous power balance reads:

$$\dot{\mathcal{E}}_t = \Pi_{\text{IN},t} - \Pi_{\text{d},t} - \Pi_{\text{rad},t},$$

as an equality of second-order random variables.

- Considering the mathematical expectation with $\mathbb{E}\{\mathcal{E}_t\} = \text{Constant}$ and $\mathbb{E}\{\dot{\mathcal{E}}_t\} = 0$:

$$\begin{aligned}\mathbb{E}\{\Pi_{\text{IN},t}\} &= \mathbb{E}\{\Pi_{\text{d},t}\} + \mathbb{E}\{\Pi_{\text{rad},t}\} \\ &= \omega_0 \eta_{\text{tot}}(\omega_0) \mathbb{E}\{\mathcal{E}_t\},\end{aligned}$$

where:

$$\eta_{\text{tot}}(\omega_0) = \eta_{\text{rad}}(\omega_0) + \eta_{\text{s}}(\omega_0) \sqrt{\frac{M_{\text{s}}}{M_{\text{s}} + \varrho_{\text{rad}}(\omega_0) |\Omega_{\text{s}}|}}$$

if ω_0 is somehow close to a "wet" eigenfrequency.

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- The power flow between two coupled sub-systems is roughly proportional to the difference of their mechanical energies, provided that:
 - 1 The sub-systems are weakly dissipative;
 - 2 Their coupling is conservative;
 - 3 They are loaded by uncorrelated "rain-on-the-roof" noises, the bandwidths of which are large with respect to the equivalent bandwidths of the sub-systems modes.
- This framework is applicable to the interaction of a structure with a cavity filled with an acoustic fluid.
- The average radiated power of a structure coupled to an acoustic fluid is proportional to its mechanical energy. This effect is characterized by an equivalent added loss factor in the energy balance.
- **Outlook:** formulation for n coupled sub-systems.

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